Supplementary Material: Low-Magnification SEM Suffices: Interpretable Deep Learning for Multi-Scale Fracture-Cause Classification in Zirconia-Toughened Alumina

Julian Schmid, Pawel Astankow, Tom Vater, Julius Beck, Robert Cichon, and Danny Krautz

CeramTec GmbH, Plochingen, Germany

This document accompanies the paper "Low-Magnification SEM Suffices..." and provides additional details on the hyperparameter search, optimal configuration, per-fold metrics, and learning curves.

Supplementary Material

Table S1. Optimal hyperparameter configuration found for the ViT model.

Parameter	**Value**
Learning Rate	7.3×10^{-5}
Weight Decay	0.140
Drop Path Rate	0.10
Label Smoothing	0.00
Warm-up Steps	103
Layer-wise LR Decay	0.843
RandAugment: #Ops	2
RandAugment: Magnitude	12
Patch Drop Rate	0.111

Table S2. Training and validation metrics (mean ± std over 5 folds).

Metric	Train Mean ± SD	Val Mean ± SD	Train CI	Val CI
Accuracy	0.998 ± 0.001	0.907 ± 0.008	[0.997, 0.999]	[0.897, 0.917]
Macro-F_1	0.998 ± 0.001	0.888 ± 0.017	[0.997, 0.999]	[0.867, 0.909]
Precision	0.998 ± 0.001	0.917 ± 0.010	[0.997, 0.999]	[0.905, 0.929]
Recall	0.998 ± 0.001	0.866 ± 0.025	[0.997, 0.999]	[0.835, 0.897]

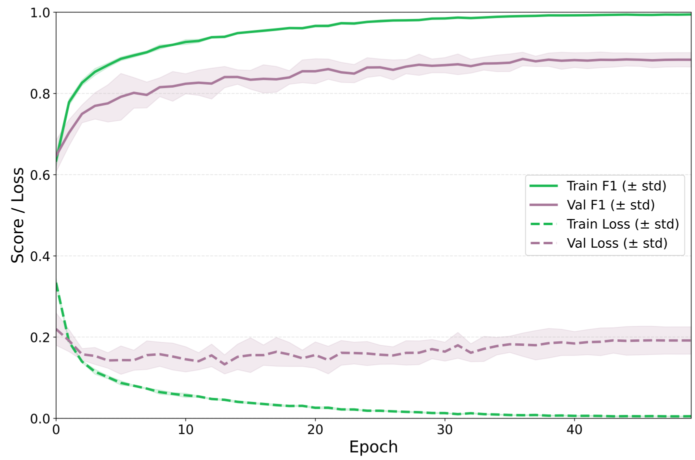

Fig. S1. Training and validation curves for macro-F_1 and loss across 50 epochs (mean ± std over 5 folds).

Table S3. ViT hyperparameter search space explored with Optuna.

Parameter	**Range / Options**	**Type**
Learning Rate	10^{-6} – 10^{-4} (log)	Continuous
Weight Decay	0.0 – 0.3	Continuous
Drop Path Rate	{0.0, 0.1, 0.2}	Categorical
Label Smoothing	{0.0, 0.05, 0.1}	Categorical
Warm-up Steps	50 – 500	Integer
Layer-wise LR Decay	0.6 – 1.0	Continuous
RandAugment Ops	1 – 3	Integer
RandAugment Magnitude	5 – 15	Integer
Patch Drop Rate	0.0 – 0.2	Continuous